

Informe de Imputación de Datos Faltantes

Dataset: World Happiness 2023

Fecha: 2026-08-20T19:06:56.268548

Dimensiones: 138 x 10

Exploración Previa

Variable	% faltantes
LGDP	6.521739130434782
Generosity	6.521739130434782
Corruption	5.072463768115942
Expectancy	2.1739130434782608
Freedom	1.4492753623188406

Test de Little (nanianr)

Campo	Valor
statistic	53.33805937618597
p_value	0.15802775601539232
conclusion	No se rechaza H0: los datos son compatibles con MCAR.

Test de Little (manual)

Campo	Valor
X2	53.34148921676224
df	44
p_value	0.15795017419885649

Estimaciones EM manuales

Campo	Valor
mu.Ladder	5.6208109510117685
mu.LGDP	9.521754918369577
mu.Support	0.7909876064977784
mu.Expectancy	65.20246195213977
mu.Freedom	0.7908977461946555
mu.Generosity	0.03202252092964119
mu.Corruption	0.7186582995550078
mu.Positive	0.6521191547314326
mu.Negative	0.2934614236912002
sigma.Ladder.Ladder	1.2890019256211198
sigma.Ladder.LGDP	1.0328045360636098
sigma.Ladder.Support	0.11610061326071636

Campo	Valor
sigma.Ladder.Expectancy	4.647093567288829
sigma.Ladder.Freedom	0.08638706650195281
sigma.Ladder.Generosity	0.013282366484845489
sigma.Ladder.Corruption	-0.09253356842300088
sigma.Ladder.Positive	0.06728703257834012
sigma.Ladder.Negative	-0.05628572708119368
sigma.LGDP.Ladder	1.0328045360636098
sigma.LGDP.LGDP	1.3430204835829276
sigma.LGDP.Support	0.11446933938039308
sigma.LGDP.Expectancy	5.470315549860604
sigma.LGDP.Freedom	0.05511049266574262
sigma.LGDP.Generosity	-0.011861779397142284
sigma.LGDP.Corruption	-0.08673006153368257
sigma.LGDP.Positive	0.03187862270397191
sigma.LGDP.Negative	-0.054117767643156706
sigma.Support.Ladder	0.11610061326071636
sigma.Support.LGDP	0.11446933938039308
sigma.Support.Support	0.016690621202791307
sigma.Support.Expectancy	0.5053234655429232
sigma.Support.Freedom	0.007050254680321209
sigma.Support.Generosity	0.0008786890365396381
sigma.Support.Corruption	-0.005848457971613163
sigma.Support.Positive	0.005471858309616056
sigma.Support.Negative	-0.007144327837527794
sigma.Expectancy.Ladder	4.647093567288829
sigma.Expectancy.LGDP	5.470315549860605
sigma.Expectancy.Support	0.5053234655429232
sigma.Expectancy.Expectancy	30.09298874996373
sigma.Expectancy.Freedom	0.24858139973908405
sigma.Expectancy.Generosity	-0.047631060016741084
sigma.Expectancy.Corruption	-0.3738804493780173
sigma.Expectancy.Positive	0.1408350704829871
sigma.Expectancy.Negative	-0.20554528612525894
sigma.Freedom.Ladder	0.08638706650195281

Campo	Valor
sigma.Freedom.LGDP	0.05511049266574262
sigma.Freedom.Support	0.007050254680321209
sigma.Freedom.Expectancy	0.24858139973908405
sigma.Freedom.Freedom	0.014406003337551097
sigma.Freedom.Generosity	0.003833685279018888
sigma.Freedom.Corruption	-0.007001705125712465
sigma.Freedom.Positive	0.008119106821437294
sigma.Freedom.Negative	-0.004657648776794533
sigma.Generosity.Ladder	0.013282366484845489
sigma.Generosity.LGDP	-0.011861779397142284
sigma.Generosity.Support	0.0008786890365396381
sigma.Generosity.Expectancy	-0.047631060016741084
sigma.Generosity.Freedom	0.003833685279018888
sigma.Generosity.Generosity	0.026096188099252238
sigma.Generosity.Corruption	-0.003764388384559922
sigma.Generosity.Positive	0.0038548335591984868
sigma.Generosity.Negative	-0.00014563191530112056
sigma.Corruption.Ladder	-0.09253356842300088
sigma.Corruption.LGDP	-0.08673006153368257
sigma.Corruption.Support	-0.005848457971613163
sigma.Corruption.Expectancy	-0.3738804493780173
sigma.Corruption.Freedom	-0.007001705125712465
sigma.Corruption.Generosity	-0.003764388384559922
sigma.Corruption.Corruption	0.031086949573787336
sigma.Corruption.Positive	-0.004437857469841132
sigma.Corruption.Negative	0.004695534606875736
sigma.Positive.Ladder	0.06728703257834012
sigma.Positive.LGDP	0.03187862270397191
sigma.Positive.Support	0.005471858309616056
sigma.Positive.Expectancy	0.1408350704829871
sigma.Positive.Freedom	0.008119106821437294
sigma.Positive.Generosity	0.0038548335591984868
sigma.Positive.Corruption	-0.004437857469841132
sigma.Positive.Positive	0.011803699114119385

Campo	Valor
sigma.Positive.Negative	-0.003838670515219225
sigma.Negative.Ladder	-0.05628572708119368
sigma.Negative.LGDP	-0.054117767643156706
sigma.Negative.Support	-0.007144327837527794
sigma.Negative.Expectancy	-0.20554528612525894
sigma.Negative.Freedom	-0.004657648776794533
sigma.Negative.Generosity	-0.00014563191530112056
sigma.Negative.Corruption	0.004695534606875736
sigma.Negative.Positive	-0.003838670515219225
sigma.Negative.Negative	0.007834365155869858

Estimaciones EM mvnmle

Campo	Valor
mu.Ladder	5.623384806821424
mu.LGDP	9.52627689836514
mu.Support	0.7913035220968396
mu.Expectancy	65.23033039570159
mu.Freedom	0.7908737994215469
mu.Generosity	0.03168512627087934
mu.Corruption	0.7182030370597277
mu.Positive	0.6519537375999197
mu.Negative	0.2934131424645525
sigma.Ladder.Ladder	1.2904348566509414
sigma.Ladder.LGDP	1.0333231639434493
sigma.Ladder.Support	0.1163481480510261
sigma.Ladder.Expectancy	4.646234203814541
sigma.Ladder.Freedom	0.08651644638634845
sigma.Ladder.Generosity	0.013124804930347744
sigma.Ladder.Corruption	-0.09258715642249152
sigma.Ladder.Positive	0.06753990100047008
sigma.Ladder.Negative	-0.05648117163097488
sigma.LGDP.Ladder	1.0333231639434493
sigma.LGDP.LGDP	1.3435386673438912
sigma.LGDP.Support	0.11471633077344098

Campo	Valor
sigma.LGDP.Expectancy	5.468670849807015
sigma.LGDP.Freedom	0.055240952501178074
sigma.LGDP.Generosity	-0.011887613269416451
sigma.LGDP.Corruption	-0.08657352980186109
sigma.LGDP.Positive	0.03196582716282158
sigma.LGDP.Negative	-0.054264761307716396
sigma.Support.Ladder	0.1163481480510261
sigma.Support.LGDP	0.11471633077344098
sigma.Support.Support	0.01673311417773083
sigma.Support.Expectancy	0.5063825693141251
sigma.Support.Freedom	0.007143301120954025
sigma.Support.Generosity	0.0008062431391659669
sigma.Support.Corruption	-0.0058876307225616655
sigma.Support.Positive	0.00557614703223714
sigma.Support.Negative	-0.007189218327220985
sigma.Expectancy.Ladder	4.646234203814541
sigma.Expectancy.LGDP	5.468670849807015
sigma.Expectancy.Support	0.5063825693141251
sigma.Expectancy.Expectancy	30.102548695520753
sigma.Expectancy.Freedom	0.2484331087348716
sigma.Expectancy.Generosity	-0.0478488005252439
sigma.Expectancy.Corruption	-0.37282479158857385
sigma.Expectancy.Positive	0.14068944159111108
sigma.Expectancy.Negative	-0.20621152828993572
sigma.Freedom.Ladder	0.08651644638634845
sigma.Freedom.LGDP	0.055240952501178074
sigma.Freedom.Support	0.007143301120954025
sigma.Freedom.Expectancy	0.2484331087348716
sigma.Freedom.Freedom	0.014394270263104997
sigma.Freedom.Generosity	0.0038054827661340653
sigma.Freedom.Corruption	-0.00701423984147696
sigma.Freedom.Positive	0.00812797943541725
sigma.Freedom.Negative	-0.004671695799528958
sigma.Generosity.Ladder	0.013124804930347744

Campo	Valor
sigma.Generosity.LGDP	-0.011887613269416451
sigma.Generosity.Support	0.0008062431391659669
sigma.Generosity.Expectancy	-0.0478488005252439
sigma.Generosity.Freedom	0.0038054827661340653
sigma.Generosity.Generosity	0.025908971625888327
sigma.Generosity.Corruption	-0.003916172300995356
sigma.Generosity.Positive	0.003765355519925349
sigma.Generosity.Negative	-8.957195507303399e-05
sigma.Corruption.Ladder	-0.09258715642249152
sigma.Corruption.LGDP	-0.08657352980186109
sigma.Corruption.Support	-0.0058876307225616655
sigma.Corruption.Expectancy	-0.37282479158857385
sigma.Corruption.Freedom	-0.00701423984147696
sigma.Corruption.Generosity	-0.003916172300995356
sigma.Corruption.Corruption	0.03123454559812288
sigma.Corruption.Positive	-0.004362951368420916
sigma.Corruption.Negative	0.004627510895910444
sigma.Positive.Ladder	0.06753990100047008
sigma.Positive.LGDP	0.03196582716282158
sigma.Positive.Support	0.00557614703223714
sigma.Positive.Expectancy	0.14068944159111108
sigma.Positive.Freedom	0.00812797943541725
sigma.Positive.Generosity	0.003765355519925349
sigma.Positive.Corruption	-0.004362951368420916
sigma.Positive.Positive	0.011863110352434064
sigma.Positive.Negative	-0.003847273770486831
sigma.Negative.Ladder	-0.05648117163097488
sigma.Negative.LGDP	-0.054264761307716396
sigma.Negative.Support	-0.007189218327220985
sigma.Negative.Expectancy	-0.20621152828993572
sigma.Negative.Freedom	-0.004671695799528958
sigma.Negative.Generosity	-8.957195507303399e-05
sigma.Negative.Corruption	0.004627510895910444
sigma.Negative.Positive	-0.003847273770486831

Campo	Valor
sigma.Negative.Negative	0.007850092704218093

Decisión del Método

Método: mice

Se eligió mice con goal=prediction, max_pct=6.52 y ratio=13.80.

Resultado de la Imputación

Campo	Valor
point_estimate.Ladder	5.62081095101177
point_estimate.LGDP	9.53061077698418
point_estimate.Support	0.790987606497778
point_estimate.Expectancy	65.2063765594925
point_estimate.Freedom	0.791011339426041
point_estimate.Generosity	0.0323055560560559
point_estimate.Corruption	0.717143141834632
point_estimate.Positive	0.652119154731433
point_estimate.Negative	0.2934614236912
severity.lambda	0.034058035543662
severity.r	0.039332662358813
severity.df	122.718535938885
severity.gamma	0.0531506585163489

Cumplimiento de Supuestos

Variable	Supuesto	Prueba/Detalle	Resultado	Cumple
Ladder	convergence	Convergencia de cadenas MICE	No convergio	False
LGDP	convergence	Convergencia de cadenas MICE	Convergio	True
Support	convergence	Convergencia de cadenas MICE	No convergio	False
Expectancy	convergence	Convergencia de cadenas MICE	No convergio	False
Freedom	convergence	Convergencia de cadenas MICE	Convergio	True
Generosity	convergence	Convergencia de cadenas MICE	Convergio	True
Corruption	convergence	Convergencia de cadenas MICE	No convergio	False
Positive	convergence	Convergencia de cadenas MICE	No convergio	False
Negative	convergence	Convergencia de cadenas MICE	No convergio	False
Ladder	distribution_comparison	Anderson-Darling (k-muestras)	stat=No aplica; p=No aplica	No aplica
LGDP	distribution_comparison	Anderson-Darling (k-muestras)	stat=1.2387; p=0.25219	True

Variable	Supuesto	Prueba/Detalle	Resultado	Cumple
Support	distribution_comparison	Anderson-Darling (k-muestras)	stat=No aplica; p=No aplica	No aplica
Expectancy	distribution_comparison	Anderson-Darling (k-muestras)	stat=1.0895; p=0.31286	True
Freedom	distribution_comparison	Anderson-Darling (k-muestras)	stat=1.3431; p=0.21805	True
Generosity	distribution_comparison	Anderson-Darling (k-muestras)	stat=0.82805; p=0.46209	True
Corruption	distribution_comparison	Anderson-Darling (k-muestras)	stat=1.8876; p=0.10478	True
Positive	distribution_comparison	Anderson-Darling (k-muestras)	stat=No aplica; p=No aplica	No aplica
Negative	distribution_comparison	Anderson-Darling (k-muestras)	stat=No aplica; p=No aplica	No aplica

Advertencias

Sin advertencias